Search strategy in EmBase:

1. distal humerus intercondylar fracture/ or intercondylar fractures of the distal humerus/ or ORIF/ or type C fractures of the distal humerus/ or intercondylar distal humeral fracture/ or humeral intercondylar fracture/
2. olecranon osteotomy/ or surgical/ or operative/ or surgery
3. 1 and 2

Supplemental Table S1. Quality assessment of randomized controlled trials by the Cochrane's tool.

Study	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Other bias
Fu 2015 [22]	Low	High	High	Low	Unclear	Unclear	Low
Khalid 2015 [23]	Low	Unclear	High	Low	Low	Unclear	Low
Liu 2017 [24]	Low	High	High	Low	Unclear	Unclear	Low
Wang 2017 [25]	Low	High	High	Low	Unclear	Unclear	Low

Supplemental Table S2. Quality assessment of non-randomized studies by the modified Newcastle-Ottawa scale.

Study	Selection				Comparability	Outcome		NOS
	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure	Demonstration that outcomes was not present at start of study	Comparability on the basis of the design or analysis	Assessment of outcome	Adequate follow-up duration	Adequate follow-up rate	Overall score
Pajarinen 2002 [18]	0	1	1	1	1	1	0	0	5
Liu 2009 [19]	0	1	1	1	1	1	0	1	6
Chen 2011 [20]	0	1	1	1	1	1	0	1	6
Zhang 2014 [21]	0	1	1	1	1	1	0	0	5
Tan 2018 [26]	0	1	1	1	1	1	0	0	5